

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY CHANGA

B.Tech. (IT) Sem - III Examination, 2010

IT - 202 Data Communication

Date: 27.11.2010, Saturday

Time: 10.30a.m to 1.30 p.m

Maximum Marks:70

Instructions:

1. Attempt all the questions.
2. Figures to the right indicate *full* marks.
3. Make suitable assumptions and neat figures if required.

SECTION- I

Q1 (a) Attempt all question

05

- (1) What is passive circuit?
- (2) What is quality factor for tuned circuit?
- (3) What is the requirement of modulation?
- (4) Give different application of voltage controlled oscillator.
- (5) Give the different type of tracking and which is best suitable?

Q1 (b) Give short note on skin effect.

06

Q2 (a) Explain the amplitude modulation in detail.

06

- (b) Explain series tuned circuit and give series Q factor, impedance in terms of Q, relative response in decibels.

06

OR

Q2 (a) Explain double sideband suppressed carrier (DSBSC) modulation

06

- (b) Short note on capacitive tap.

06

Q3 (a) Three resistors have values $R_1=10\text{Kohm}$, $R_2=14\text{Kohm}$ and $R_3=24\text{Kohm}$. It is known that the thermal noise voltage generated by R_1 is 0.3Microvolt. calculate thermal noise voltage generated by (a) the Three resistors connected in series (b) connected in parallel.

06

- (b) Explain mutual inductance with suitable equation of series and parallel coil, mutual inductance with minimum and maximum value.

06

OR

Q3 (a) At frequencies well below the self resonant frequency, a coil has a Series resistance of 10 ohm and an inductance of 15 microhenry.

P.T.O.



The self capacitance is 50 picofarad. The coil is used as inductance in series tuned circuit at a frequency of 1.3Mhz. Determine the effective inductance and the effective Qfactor.

06

(b) Draw and explain the amplitude modulated transmitters.

06

SECTION - II

FOR REFERENCE

Q4 (a) Give the different sources of noise

05

(b) Three amplifiers 1, 2 and 3 have the following characteristics.

$F_1=9$ db, $G_1=48$ db

$F_2=6$ db, $G_2=35$ db

$F_3=4$ db, $G_3=20$ db

06

The amplifiers are connected in tandem. Determine which combination Gives the lowest noise factor referred to the input

Q5 (a) Explain the power signal. Give one example each of deterministic and nondeterministic power signal.

06

(b) Explain Voltage control oscillator with clapp oscillator & Operational amplifier (op-amp). Give necessary equation related to it.

06

OR

Q5 (a) A carrier wave of frequency 18 MHz and peak value 15V is amplitude modulated by a 13 KHz sine wave of amplitude 10 V. determine the modulation index and draw the amplitude spectrum

06

(b) Explain amplitude demodulator circuits

06

Q6 (a) Explain block diagram of superhetrodyne receiver

06

(b) Explain gain and sensitivity of receiver in detail.

06

OR

Q6 (a) A high-frequency transformer has identical primary and secondary Circuit for which $L_p=L_s=150$ microhenry, $C_p=C_s=470$ Pico farad and the Q factor for each circuit alone (that is, not coupled) is 85. The coefficient of coupling is 0.01. The load resistance is 5000 ohm, and the constant current source feeding the transformer has an internal resistance of 75 Kohm. Determine the transfer impedance at resonance.

06

(b) A receiver tunes signals from 550 to 1600 KHz with an IF of 455 KHz. Find the frequency tuning ranges and capacitor tuning ranges for the oscillator section and for the RF section.

06